

Product, Quotient, and Trigonometric Derivatives

ANSWER KEY



The product rule

- 1 Differentiate $f(x) = (3x^2 - 1)(x^3 + 2x)$ using the product rule.

$$f'(x) = 6x(x^3 + 2x) + (3x^2 - 1)(3x^2 + 2) = 6x^4 + 12x^2 + 9x^4 + 6x^2 - 3x^2 - 2 = 15x^4 + 15x^2 - 2.$$

- 2 Differentiate $f(x) = x^2 \sin x$.

$$f'(x) = 2x \sin x + x^2 \cos x.$$

- 3 Differentiate $f(x) = e^x \cos x$.

$$f'(x) = e^x \cos x - e^x \sin x = e^x (\cos x - \sin x).$$

The quotient rule

- 4 Differentiate $f(x) = \frac{3x-1}{x+4}$.

$$f'(x) = \frac{3(x+4) - (3x-1)(1)}{(x+4)^2} = \frac{3x+12-3x+1}{(x+4)^2} = \frac{13}{(x+4)^2}.$$

- 5 Differentiate $f(x) = \frac{x^2}{x^2+1}$.

$$f'(x) = \frac{2x(x^2+1) - x^2(2x)}{(x^2+1)^2} = \frac{2x^3+2x-2x^3}{(x^2+1)^2} = \frac{2x}{(x^2+1)^2}.$$

- 6 Differentiate $f(x) = \frac{\sin x}{x}$.

$$f'(x) = \frac{x \cos x - \sin x}{x^2}.$$

Derivatives of the six trig functions

- 7 State the derivative of each trig function:

a $\frac{d}{dx} \sin x = \cos x$

b $\frac{d}{dx} \cos x = -\sin x$

c $\frac{d}{dx} \tan x = \sec^2 x$

d $\frac{d}{dx} \cot x = -\csc^2 x$

e $\frac{d}{dx} \sec x = \sec x \tan x$

f $\frac{d}{dx} \csc x = -\csc x \cot x$

- 8 Differentiate $f(x) = x \tan x$.

$$f'(x) = \tan x + x \sec^2 x.$$

- 9 Differentiate $f(x) = \sec x \cdot \csc x$.

$$f'(x) = \sec x \tan x \csc x + \sec x (-\csc x \cot x) = \sec x \csc x (\tan x - \cot x).$$

Combining product, quotient, and trig rules

- 10 Differentiate $f(x) = \frac{e^x}{\sin x}$.

$$f'(x) = \frac{e^x \sin x - e^x \cos x}{\sin^2 x} = \frac{e^x(\sin x - \cos x)}{\sin^2 x}.$$

- 11 Find the equation of the tangent line to $f(x) = x \cos x$ at $x = 0$.

$$f(0) = 0. f'(x) = \cos x - x \sin x; f'(0) = 1. \text{ Tangent: } y - 0 = 1(x - 0), \text{ i.e. } y = x.$$

Derivatives using a table of values

- 12 Given $f(3) = 2$, $f'(3) = -1$, $g(3) = 4$, $g'(3) = 5$, find $h'(3)$ if $h(x) = f(x)g(x)$.

$$h'(x) = f'(x)g(x) + f(x)g'(x); h'(3) = (-1)(4) + (2)(5) = -4 + 10 = 6.$$

- 13 Using the same table, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}; k'(3) = \frac{(-1)(4) - (2)(5)}{4^2} = \frac{-4 - 10}{16} = -\frac{14}{16} = -\frac{7}{8}.$$

Second derivatives with product and quotient rules

- 14 Find $f''(x)$ for $f(x) = x \sin x$.

$$f'(x) = \sin x + x \cos x. f''(x) = \cos x + \cos x - x \sin x = 2 \cos x - x \sin x.$$

More product and quotient rule practice

- 15 Differentiate $f(x) = (2x + 1)^2(x - 3)$, treating $(2x + 1)^2$ as one factor.

$$\text{Let } u = (2x + 1)^2, u' = 2(2x + 1)(2) = 4(2x + 1):$$

$$f'(x) = 4(2x + 1)(x - 3) + (2x + 1)^2(1) = (2x + 1)[4(x - 3) + (2x + 1)] = (2x + 1)(6x - 11).$$

- 16 Differentiate $f(x) = \frac{x^2 + 1}{x - 1}$.

$$f'(x) = \frac{2x(x - 1) - (x^2 + 1)(1)}{(x - 1)^2} = \frac{2x^2 - 2x - x^2 - 1}{(x - 1)^2} = \frac{x^2 - 2x - 1}{(x - 1)^2}.$$

An applied product-rule problem

- 17 A company's revenue is $R(t) = p(t) \cdot q(t)$, where $p(t)$ is price and $q(t)$ is quantity sold, both functions of time t (months). At $t = 6$, $p(6) = 20$, $p'(6) = 0.5$, $q(6) = 100$, $q'(6) = -2$. Find $R'(6)$ and interpret its sign.

$R'(t) = p'(t)q(t) + p(t)q'(t)$: $R'(6) = (0.5)(100) + (20)(-2) = 50 - 40 = 10$. Since $R'(6) > 0$, revenue is increasing at $t = 6$ despite falling quantity, because the price increase outweighs the quantity decrease.

Higher-order trigonometric derivatives

- 18 Find $f''(x)$ for $f(x) = \sin x$, and describe the repeating pattern of derivatives f, f', f'', f''', f'''' .

$f'(x) = \cos x$, $f''(x) = -\sin x$. The pattern $\sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \sin x$ repeats every 4 derivatives.

- 19 Using the repeating pattern above, find $f^{(50)}(x)$ for $f(x) = \sin x$ (the 50th derivative).

Since the pattern has period 4, and $50 = 4(12) + 2$, the 50th derivative matches the 2nd derivative in the cycle: $f^{(50)}(x) = -\sin x$.

A capstone combined-rule problem

- 20 Differentiate $f(x) = \frac{x^2 \sin x}{x+1}$, applying the quotient rule with the product rule for the numerator's derivative.

Numerator $u = x^2 \sin x$: $u' = 2x \sin x + x^2 \cos x$. Denominator $v = x + 1$: $v' = 1$.
 $f'(x) = \frac{(2x \sin x + x^2 \cos x)(x+1) - x^2 \sin x(1)}{(x+1)^2}$.

- 21 An object's position is $s(t) = t^2 \cos t$ (meters, seconds). Find the object's velocity function $s'(t)$ using the product rule, then find its velocity at $t = 0$.

$s'(t) = 2t \cos t - t^2 \sin t$. At $t = 0$: $s'(0) = 2(0)(1) - 0 = 0$ m/s (momentarily at rest).

- 22 Differentiate $f(x) = \frac{\cos x}{1 + \sin x}$, and simplify your answer using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$.

$f'(x) = \frac{-\sin x(1 + \sin x) - \cos x(\cos x)}{(1 + \sin x)^2} = \frac{-\sin x - \sin^2 x - \cos^2 x}{(1 + \sin x)^2} = \frac{-\sin x - 1}{(1 + \sin x)^2} = \frac{-1}{1 + \sin x}$.

- 23 Given $f(2) = 5$, $f'(2) = 3$, $g(2) = -1$, $g'(2) = 4$, find each of the following at $x = 2$:

a $\frac{d}{dx}[f(x)g(x)]$ $f'(2)g(2) + f(2)g'(2) = (3)(-1) + (5)(4) = -3 + 20 = 17$.

b $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]$ $\frac{f'(2)g(2) - f(2)g'(2)}{[g(2)]^2} = \frac{(3)(-1) - (5)(4)}{1} = \frac{-3 - 20}{1} = -23$.

c $\frac{d}{dx}[3f(x) - 2g(x)]$ $3f'(2) - 2g'(2) = 3(3) - 2(4) = 9 - 8 = 1$.

Multiple choice — easy 15

24 Given $f(2) = 3$, $f'(2) = 5$, $g(2) = 4$, $g'(2) = -1$, find $h'(2)$ if $h(x) = f(x)g(x)$.

- a. -5
- b. 23
- c. 17 ✓
- d. 12

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(2) = (5)(4) + (3)(-1) = 20 - 3 = 17.$$

25 Given $f(7) = 4$, $f'(7) = -4$, $g(7) = 6$, $g'(7) = -2$, find $h'(7)$ if $h(x) = f(x)g(x)$.

- a. -16
- b. 24
- c. 8
- d. -32 ✓

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(7) = (-4)(6) + (4)(-2) = -24 - 8 = -32.$$

26 Given $f(6) = -5$, $f'(6) = 1$, $g(6) = 1$, $g'(6) = -1$, find $h'(6)$ if $h(x) = f(x)g(x)$.

- a. -4
- b. 6 ✓
- c. -1
- d. -5

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(6) = (1)(1) + (-5)(-1) = 1 + 5 = 6.$$

27 Given $f(4) = 5$, $f'(4) = -4$, $g(4) = 1$, $g'(4) = 5$, find $h'(4)$ if $h(x) = f(x)g(x)$.

- a. -20
- b. 5
- c. -29
- d. 21 ✓

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(4) = (-4)(1) + (5)(5) = -4 + 25 = 21.$$

28 Given $f(4) = -3$, $f'(4) = 3$, $g(4) = -5$, $g'(4) = 1$, find $h'(4)$ if $h(x) = f(x)g(x)$.

a. -12

b. 15

c. 3

d. -18 ✓

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(4) = (3)(-5) + (-3)(1) = -15 - 3 = -18.$$

29 Given $f(5) = 3$, $f'(5) = 4$, $g(5) = -6$, $g'(5) = -1$, find $h'(5)$ if $h(x) = f(x)g(x)$.

a. -4

b. -21

c. -18

d. -27 ✓

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(5) = (4)(-6) + (3)(-1) = -24 - 3 = -27.$$

30 Given $f(3) = -4$, $f'(3) = 6$, $g(3) = 3$, $g'(3) = -4$, find $h'(3)$ if $h(x) = f(x)g(x)$.

a. -24

b. -12

c. 2

d. 34 ✓

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(3) = (6)(3) + (-4)(-4) = 18 + 16 = 34.$$

31 Given $f(7) = 4$, $f'(7) = -2$, $g(7) = -5$, $g'(7) = 1$, find $h'(7)$ if $h(x) = f(x)g(x)$.

a. 14 ✓

b. -2

c. -20

d. 6

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(7) = (-2)(-5) + (4)(1) = 10 + 4 = 14.$$

32 Given $f(3) = 2$, $f'(3) = 4$, $g(3) = -5$, $g'(3) = 3$, find $h'(3)$ if $h(x) = f(x)g(x)$.

a. -10

b. -14 ✓

c. -26

d. 12

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(3) = (4)(-5) + (2)(3) = -20 + 6 = -14.$$

33 Given $f(5) = 2$, $f'(5) = -2$, $g(5) = 5$, $g'(5) = -2$, find $h'(5)$ if $h(x) = f(x)g(x)$.

a. -14 ✓

b. 4

c. 10

d. -6

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(5) = (-2)(5) + (2)(-2) = -10 - 4 = -14.$$

34 Given $f(6) = 3$, $f'(6) = -1$, $g(6) = -6$, $g'(6) = 1$, find $h'(6)$ if $h(x) = f(x)g(x)$.

a. -1

b. 9 ✓

c. 3

d. -18

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(6) = (-1)(-6) + (3)(1) = 6 + 3 = 9.$$

35 Given $f(4) = 6$, $f'(4) = -4$, $g(4) = 6$, $g'(4) = 5$, find $h'(4)$ if $h(x) = f(x)g(x)$.

a. -54

b. -20

c. 36

d. 6 ✓

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(4) = (-4)(6) + (6)(5) = -24 + 30 = 6.$$

36 Given $f(8) = 1$, $f'(8) = -3$, $g(8) = -3$, $g'(8) = -5$, find $h'(8)$ if $h(x) = f(x)g(x)$.

a. 14

b. 4 ✓

c. 15

d. -3

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(8) = (-3)(-3) + (1)(-5) = 9 - 5 = 4.$$

37 Given $f(3) = -2$, $f'(3) = -3$, $g(3) = -4$, $g'(3) = -1$, find $h'(3)$ if $h(x) = f(x)g(x)$.

a. 3

b. 10

c. 14 ✓

d. 8

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(3) = (-3)(-4) + (-2)(-1) = 12 + 2 = 14.$$

38 Given $f(4) = -5$, $f'(4) = 1$, $g(4) = 4$, $g'(4) = 6$, find $h'(4)$ if $h(x) = f(x)g(x)$.

a. -26 ✓

b. 34

c. -20

d. 6

$$h'(x) = f'(x)g(x) + f(x)g'(x), \text{ so } h'(4) = (1)(4) + (-5)(6) = 4 - 30 = -26.$$

Multiple choice — medium 15

39 Given $f(2) = -5$, $f'(2) = -4$, $g(2) = 5$, $g'(2) = -4$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{8}{5}$

b. -8

c. 0

d. $-\frac{8}{5}$ ✓

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(2) = \frac{(-4)(5) - (-5)(-4)}{(5)^2} = \frac{-40}{25} = -\frac{8}{5}.$$

40 Given $f(4) = -4$, $f'(4) = -1$, $g(4) = 6$, $g'(4) = -2$, find $k'(4)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{1}{18}$

b. $\frac{7}{18}$

c. $-\frac{7}{18}$ ✓

d. $-\frac{7}{3}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(4) = \frac{(-1)(6) - (-4)(-2)}{(6)^2} = \frac{-14}{36} = -\frac{7}{18}.$$

41 Given $f(2) = -1$, $f'(2) = 6$, $g(2) = -3$, $g'(2) = 4$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{14}{3}$

b. $-\frac{22}{9}$

c. $-\frac{14}{9}$ ✓

d. $\frac{14}{9}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(2) = \frac{(6)(-3) - (-1)(4)}{(-3)^2} = \frac{-14}{9} = -\frac{14}{9}.$$

42 Given $f(2) = 6$, $f'(2) = -1$, $g(2) = 5$, $g'(2) = -6$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{31}{25}$ ✓

b. $-\frac{41}{25}$

c. $-\frac{31}{25}$

d. $\frac{31}{5}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(2) = \frac{(-1)(5) - (6)(-6)}{(5)^2} = \frac{31}{25} = \frac{31}{25}.$$

43 Given $f(4) = -2$, $f'(4) = 2$, $g(4) = -3$, $g'(4) = -1$, find $k'(4)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{4}{9}$

b. $\frac{8}{9}$

c. $\frac{8}{3}$

d. $-\frac{8}{9}$ ✓

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(4) = \frac{(2)(-3) - (-2)(-1)}{(-3)^2} = \frac{-8}{9} = -\frac{8}{9}.$$

44 Given $f(2) = -1$, $f'(2) = -1$, $g(2) = -2$, $g'(2) = 2$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. -2

b. -1

c. 0

d. 1 ✓

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(2) = \frac{(-1)(-2) - (-1)(2)}{(-2)^2} = \frac{4}{4} = 1.$$

45 Given $f(5) = 2$, $f'(5) = 2$, $g(5) = 2$, $g'(5) = -3$, find $k'(5)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{5}{2}$

b. $\frac{5}{2}$ ✓

c. $-\frac{1}{2}$

d. 5

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(5) = \frac{(2)(2) - (2)(-3)}{(2)^2} = \frac{10}{4} = \frac{5}{2}.$$

46 Given $f(4) = -6$, $f'(4) = 4$, $g(4) = -4$, $g'(4) = 2$, find $k'(4)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{1}{4}$

b. $-\frac{7}{4}$

c. $-\frac{1}{4}$ ✓

d. 1

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(4) = \frac{(4)(-4) - (-6)(2)}{(-4)^2} = \frac{-4}{16} = -\frac{1}{4}.$$

47 Given $f(2) = -3$, $f'(2) = 4$, $g(2) = 2$, $g'(2) = 4$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. -1

b. -5

c. 5 ✓

d. 10

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(2) = \frac{(4)(2) - (-3)(4)}{(2)^2} = \frac{20}{4} = 5.$$

48 Given $f(3) = 4$, $f'(3) = 5$, $g(3) = 3$, $g'(3) = -1$, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{19}{9}$

b. $\frac{19}{3}$

c. $\frac{19}{9}$ ✓

d. $\frac{11}{9}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(3) = \frac{(5)(3) - (4)(-1)}{(3)^2} = \frac{19}{9} = \frac{19}{9}.$$

49 Given $f(8) = -2$, $f'(8) = -5$, $g(8) = 4$, $g'(8) = -5$, find $k'(8)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{15}{2}$

b. $-\frac{15}{8}$ ✓

c. $-\frac{5}{8}$

d. $\frac{15}{8}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(8) = \frac{(-5)(4) - (-2)(-5)}{(4)^2} = \frac{-30}{16} = -\frac{15}{8}.$$

50 Given $f(3) = -6$, $f'(3) = 2$, $g(3) = -4$, $g'(3) = 1$, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{7}{8}$

b. $\frac{1}{8}$

c. $-\frac{1}{8}$ ✓

d. $\frac{1}{2}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(3) = \frac{(2)(-4) - (-6)(1)}{(-4)^2} = \frac{-2}{16} = -\frac{1}{8}.$$

51 Given $f(6) = 3$, $f'(6) = -2$, $g(6) = 4$, $g'(6) = 6$, find $k'(6)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{5}{8}$

b. $-\frac{13}{2}$

c. $\frac{13}{8}$

d. $-\frac{13}{8}$ ✓

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(6) = \frac{(-2)(4) - (3)(6)}{(4)^2} = \frac{-26}{16} = -\frac{13}{8}.$$

52 Given $f(3) = -6$, $f'(3) = -6$, $g(3) = 6$, $g'(3) = -3$, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. -9

b. $\frac{3}{2}$

c. $-\frac{3}{2}$ ✓

d. $-\frac{1}{2}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(3) = \frac{(-6)(6) - (-6)(-3)}{(6)^2} = \frac{-54}{36} = -\frac{3}{2}.$$

53 Given $f(8) = -5$, $f'(8) = 5$, $g(8) = 4$, $g'(8) = 6$, find $k'(8)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{25}{2}$

b. $\frac{25}{8}$ ✓

c. $-\frac{5}{8}$

d. $-\frac{25}{8}$

$$k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \text{ so } k'(8) = \frac{(5)(4) - (-5)(6)}{(4)^2} = \frac{50}{16} = \frac{25}{8}.$$

Multiple choice — hard 15

54 Use the product rule to differentiate $f(x) = (2x - 5)(4x - 4)$, then find $f'(1)$.

a. -12 ✓

b. 0

c. 8

d. -10

$$f'(x) = 2(4x - 4) + 4(2x - 5), \text{ so } f'(1) = 2(0) + 4(-3) = 0 - 12 = -12.$$

55 Use the product rule to differentiate $f(x) = (4x - 2)(5x - 4)$, then find $f'(1)$.

- a. 2
- b. 16
- c. 14 ✓
- d. 20

$$f'(x) = 4(5x - 4) + 5(4x - 2), \text{ so } f'(1) = 4(1) + 5(2) = 4 + 10 = 14.$$

56 Use the product rule to differentiate $f(x) = (5x - 3)(4x + 2)$, then find $f'(-1)$.

- a. -40
- b. -42 ✓
- c. 20
- d. 16

$$f'(x) = 5(4x + 2) + 4(5x - 3), \text{ so } f'(-1) = 5(-2) + 4(-8) = -10 - 32 = -42.$$

57 Use the product rule to differentiate $f(x) = (x - 2)(2x - 3)$, then find $f'(3)$.

- a. 3
- b. 5 ✓
- c. 7
- d. 2

$$f'(x) = 1(2x - 3) + 2(x - 2), \text{ so } f'(3) = 1(3) + 2(1) = 3 + 2 = 5.$$

58 Use the product rule to differentiate $f(x) = (2x + 1)(2x + 6)$, then find $f'(-3)$.

- a. -8
- b. -10 ✓
- c. 0
- d. 4

$$f'(x) = 2(2x + 6) + 2(2x + 1), \text{ so } f'(-3) = 2(0) + 2(-5) = 0 - 10 = -10.$$

59 Use the quotient rule to differentiate $f(x) = \frac{3x+3}{3x+6}$, then find $f'(0)$.

a. $\frac{1}{4}$ ✓

b. $-\frac{1}{4}$

c. $\frac{5}{4}$

d. $\frac{3}{2}$

$$f'(x) = \frac{3(3x+6) - 3(3x+3)}{(3x+6)^2} = \frac{9}{(3x+6)^2}, \text{ so } f'(0) = \frac{9}{36} = \frac{1}{4}.$$

60 Use the quotient rule to differentiate $f(x) = \frac{2x-6}{2x-3}$, then find $f'(0)$.

a. -2

b. $\frac{1}{6}$

c. $-\frac{2}{3}$

d. $\frac{2}{3}$ ✓

$$f'(x) = \frac{2(2x-3) - 2(2x-6)}{(2x-3)^2} = \frac{6}{(2x-3)^2}, \text{ so } f'(0) = \frac{6}{9} = \frac{2}{3}.$$

61 Use the quotient rule to differentiate $f(x) = \frac{2x-6}{3x+6}$, then find $f'(1)$.

a. $\frac{10}{27}$ ✓

b. $-\frac{7}{54}$

c. $\frac{10}{3}$

d. $-\frac{10}{27}$

$$f'(x) = \frac{2(3x+6) - 3(2x-6)}{(3x+6)^2} = \frac{30}{(3x+6)^2}, \text{ so } f'(1) = \frac{30}{81} = \frac{10}{27}.$$

62 Use the quotient rule to differentiate $f(x) = \frac{3x-1}{2x+1}$, then find $f'(2)$.

a. $\frac{1}{5}$ ✓

b. 1

c. $-\frac{1}{5}$

d. $\frac{6}{5}$

$$f'(x) = \frac{3(2x+1) - 2(3x-1)}{(2x+1)^2} = \frac{5}{(2x+1)^2}, \text{ so } f'(2) = \frac{5}{25} = \frac{1}{5}.$$

63 Use the quotient rule to differentiate $f(x) = \frac{3x-5}{3x-1}$, then find $f'(2)$.

a. $\frac{37}{25}$

b. $\frac{12}{5}$

c. $-\frac{12}{25}$

d. $\frac{12}{25}$ ✓

$$f'(x) = \frac{3(3x-1) - 3(3x-5)}{(3x-1)^2} = \frac{12}{(3x-1)^2}, \text{ so } f'(2) = \frac{12}{25} = \frac{12}{25}.$$

64 Use the product rule to differentiate $f(x) = -7x \cos x$, then find $f'(0)$.

a. -8

b. 7

c. -6

d. -7 ✓

$$f'(x) = -7 \cos x + 7x \sin x, \text{ so } f'(0) = (-7) \cos(0) - (-7)(0) \sin(0) = (-7)(1) - 0 = -7.$$

65 Use the product rule to differentiate $f(x) = -4x \cos x$, then find $f'(0)$.

a. 4

b. -5

c. -4 ✓

d. -3

$$f'(x) = -4 \cos x + 4x \sin x, \text{ so } f'(0) = (-4) \cos(0) - (-4)(0) \sin(0) = (-4)(1) - 0 = -4.$$

66 Use the quotient rule to differentiate $f(x) = \frac{\sin x}{x+2}$, then find $f'(0)$.

a. $-\frac{1}{2}$

b. $\frac{1}{3}$

c. 2

d. $\frac{1}{2}$ ✓

$$f'(x) = \frac{\cos x(x+2) - \sin x(1)}{(x+2)^2}, \text{ so } f'(0) = \frac{(1)(2) - (0)(1)}{(2)^2} = \frac{2}{4} = \frac{1}{2}.$$

67 Use the quotient rule to differentiate $f(x) = \frac{\sin x}{x-2}$, then find $f'(0)$.

a. -1

b. $\frac{1}{2}$

c. $-\frac{1}{2}$ ✓

d. -2

$$f'(x) = \frac{\cos x(x-2) - \sin x(1)}{(x-2)^2}, \text{ so } f'(0) = \frac{(1)(-2) - (0)(1)}{(-2)^2} = \frac{-2}{4} = -\frac{1}{2}.$$

68 Use the quotient rule to differentiate $f(x) = \frac{\sin x}{x-1}$, then find $f'(0)$.

a. $\frac{1}{2}$

b. 0

c. 1

d. -1 ✓

$$f'(x) = \frac{\cos x(x-1) - \sin x(1)}{(x-1)^2}, \text{ so } f'(0) = \frac{(1)(-1) - (0)(1)}{(-1)^2} = \frac{-1}{1} = -1.$$